

Network design practice 4-1

김경민

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1 Assignment 1

```
cellular@ubuntu:~/UERANSIM/config$ vi custom-gnb.yaml
cellular@ubuntu:~/UERANSIM/config$ vi custom-gnb.yaml
cellular@ubuntu:~/UERANSIM/config$ cd ..
cellular@ubuntu:~/UERANSIM$ cd build/
cellular@ubuntu:~/UERANSIM/build$ ./nr-gnb -c ../config/custom-gnb.yaml
UERANSIM v3.2.7
[2026-05-20 20:10:38.900] [sctp] [info] Trying to establish SCTP connection... (192.168.48.131:38412)
[2026-05-20 20:10:38.911] [sctp] [info] SCTP connection established (192.168.48.131:38412)
[2026-05-20 20:10:38.912] [sctp] [debug] SCTP association setup ascId[3]
[2026-05-20 20:10:38.913] [ngap] [debug] Sending NG Setup Request
[2026-05-20 20:10:38.934] [ngap] [debug] NG Setup Response received
[2026-05-20 20:10:38.934] [ngap] [info] NG Setup procedure is successful
```

Successful NG setup.

sctp				
Time	Source	Destination	Protocol	Length/Info
69.6.907754941	127.0.0.2	127.0.0.8	SCTP	108 HEARTBEAT
70.6.907893322	127.0.0.8	127.0.0.2	SCTP	108 HEARTBEAT_ACK
71.7.107904905	192.168.48.131	192.168.48.130	SCTP	108 HEARTBEAT
72.7.108047811	192.168.48.130	192.168.48.131	SCTP	108 HEARTBEAT_ACK
73.7.108150582	127.0.0.4	127.0.0.9	SCTP	108 HEARTBEAT
74.7.108080808	127.0.0.9	127.0.0.4	SCTP	108 HEARTBEAT_ACK
75.7.424172925	127.0.0.9	127.0.0.4	SCTP	108 HEARTBEAT
76.7.424207120	127.0.0.4	127.0.0.9	SCTP	108 HEARTBEAT_ACK
112.9.303452061	192.168.48.130	192.168.48.131	SCTP	108 HEARTBEAT
113.9.303515372	192.168.48.131	192.168.48.130	SCTP	108 HEARTBEAT_ACK
124.9.715539371	127.0.0.9	127.0.0.4	DIAMETER	140 cmd=Device-Watchdog Request(280) flags=R--- appl=Diameter Com...
125.9.716605101	127.0.0.4	127.0.0.9	DIAMETER	168 SACK cmd=Device-Watchdog Answer(280) flags=---- appl=Diameter...
126.9.736807741	127.0.0.8	127.0.0.2	DIAMETER	140 cmd=Device-Watchdog Request(280) flags=R--- appl=Diameter Com...
127.9.737335070	127.0.0.2	127.0.0.8	DIAMETER	168 SACK cmd=Device-Watchdog Answer(280) flags=---- appl=Diameter...
128.9.919778140	127.0.0.9	127.0.0.4	SCTP	64 SACK
129.9.944046665	127.0.0.8	127.0.0.2	SCTP	64 SACK
163.13.563064092	192.168.48.131	192.168.48.130	SCTP	108 HEARTBEAT
164.13.564304098	192.168.48.130	192.168.48.131	SCTP	108 HEARTBEAT_ACK
190.16.219984210	127.0.0.9	127.0.0.4	SCTP	108 HEARTBEAT
191.16.220104088	127.0.0.4	127.0.0.9	SCTP	108 HEARTBEAT_ACK
192.16.379688330	127.0.0.4	127.0.0.9	SCTP	108 HEARTBEAT
193.16.379709826	127.0.0.2	127.0.0.8	SCTP	108 HEARTBEAT
194.16.379839403	127.0.0.9	127.0.0.4	SCTP	108 HEARTBEAT_ACK
195.16.379867797	127.0.0.8	127.0.0.2	SCTP	108 HEARTBEAT_ACK
196.16.380100956	127.0.0.8	127.0.0.2	SCTP	108 HEARTBEAT

Frame 1: 108 bytes on wire (864 bits). 108 bytes captured (864 bits) on interface any. Id 0

Wireshark running on 5G_Core VM.

2 Assignment 2

```
cellularguinea:~#UERANSIM/build$ sudo ./nr-ue -c ../config/custom-ue.yaml
[sudo] password for cellularguinea:
UERANSIM v3.2.7
[2026-05-20 23:25:48.522] [nas] [info] UE switches to state [MM-DEREGISTERED/PLMN-SEARCH]
[2026-05-20 23:25:48.523] [rrc] [debug] New signal detected for cell[1], total [1] cells in coverage
[2026-05-20 23:25:50.723] [nas] [error] PLMN selection failure, no cells in coverage
[2026-05-20 23:25:52.922] [nas] [error] PLMN selection failure, no cells in coverage
[2026-05-20 23:25:53.523] [rrc] [warning] Acceptable cell selection failed in [1] cells. [1] no SI, [0] reserved, [0] barred, fatal [0]
[2026-05-20 23:25:53.523] [rrc] [error] Cell selection failure, no suitable or acceptable cell found
[2026-05-20 23:25:54.020] [nas] [info] UE switches to state [MM-DEREGISTERED/NO-CELL-AVAILABLE]
[2026-05-20 23:25:55.313] [nas] [info] Selected plmn[999/70]
[2026-05-20 23:25:55.313] [rrc] [info] Selected cell plmn[999/70] tac[1] category[SUITABLE]
[2026-05-20 23:25:55.313] [nas] [info] UE switches to state [MM-DEREGISTERED/PS]
[2026-05-20 23:25:55.313] [nas] [info] UE switches to state [MM-DEREGISTERED/NORMAL-SERVICE]
[2026-05-20 23:25:55.313] [nas] [debug] Initial registration required due to [MM-DEREG-NORMAL-SERVICE]
[2026-05-20 23:25:55.313] [nas] [debug] UAC access attempt is allowed for Identity[0], category[MO_sig]
[2026-05-20 23:25:55.313] [nas] [debug] Sending Initial Registration
[2026-05-20 23:25:55.325] [nas] [info] UE switches to state [MM-REGISTER-INITIATED]
[2026-05-20 23:25:55.325] [rrc] [debug] Sending RRC Setup Request
[2026-05-20 23:25:55.326] [rrc] [info] RRC connection established
[2026-05-20 23:25:55.326] [rrc] [info] UE switches to state [RRC-CONNECTED]
[2026-05-20 23:25:55.365] [nas] [debug] Authentication Request received
[2026-05-20 23:25:55.365] [nas] [debug] Received SQN [0000000000000]
[2026-05-20 23:25:55.365] [nas] [debug] SQN-MS [0000000000000]
[2026-05-20 23:25:55.365] [nas] [debug] Sending Authentication Failure due to SQN out of range
[2026-05-20 23:25:55.381] [nas] [debug] Authentication Request received
[2026-05-20 23:25:55.381] [nas] [debug] Received SQN [0000000000021]
[2026-05-20 23:25:55.381] [nas] [debug] SQN-MS [0000000000000]
[2026-05-20 23:25:55.393] [nas] [debug] Security Mode Command received
[2026-05-20 23:25:55.393] [nas] [debug] Selected Integrity[2] ciphering[0]
[2026-05-20 23:25:55.427] [nas] [debug] Registration accept received
[2026-05-20 23:25:55.427] [nas] [info] UE switches to state [MM-REGISTERED/NORMAL-SERVICE]
[2026-05-20 23:25:55.427] [nas] [debug] Sending Registration Complete
[2026-05-20 23:25:55.427] [nas] [info] Initial Registration is successful
[2026-05-20 23:25:55.427] [nas] [debug] Sending PDU Session Establishment Request
[2026-05-20 23:25:55.427] [nas] [debug] UAC access attempt is allowed for Identity[0], category[MO_sig]
[2026-05-20 23:25:55.629] [nas] [debug] Configuration Update Command received
[2026-05-20 23:25:55.681] [nas] [debug] PDU Session Establishment Accept received
[2026-05-20 23:25:55.681] [nas] [info] PDU Session establishment is successful PSI[1]
[2026-05-20 23:25:55.771] [app] [info] Connection setup for PDU session[1] is successful, TUN interface[uesimtun0, 10.45.0.2] is up.
```

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```
[2026-05-20 23:16:35.273] [scip] [debug] SCIP association Setup ascid[5]
[2026-05-20 23:16:35.273] [ngap] [debug] Sending NG Setup Request
[2026-05-20 23:16:35.352] [ngap] [debug] NG Setup Response received
[2026-05-20 23:16:35.352] [ngap] [info] NG Setup procedure is successful
ping -l uesimtun0 korea.ac.kr
[2026-05-20 23:25:48.522] [rrc] [debug] UE[1] new signal detected
[2026-05-20 23:25:55.325] [rrc] [info] RRC Setup for UE[1]
[2026-05-20 23:25:55.327] [ngap] [debug] Initial NAS message received from UE[1]
[2026-05-20 23:25:55.426] [ngap] [debug] Initial Context Setup Request received
[2026-05-20 23:25:55.680] [ngap] [info] PDU session resource(s) setup for UE[1] count[1]
```

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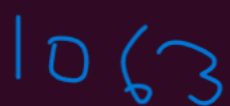
Ping -I uesimtun0 korea.ac.kr testing UE uplink traffic.



3

3 Explaining code

```
13     server:
14         - address: 127.0.0.5
15         port: 7777
16     client:
17         nrf:
18             - uri: http://127.0.0.10:7777
19     #     scp:
20     #         - uri: http://127.0.0.200:7777
21     ngap:
22         server:
23             - address: 192.168.48.131
24         metrics:
25             server:
26                 - address: 127.0.0.5
27                 port: 9090
28         guami:
29             - plmn_id:
30                 mcc: 999
31                 mnc: 70
32                 amf_id:
```



ngap server address and gtpu server address are both set to the ens33 IP address of 5G_Core VM, 192.168.48.131.

```
mcc: '999'      # Mobile Country Code value
mnc: '70'       # Mobile Network Code value (2 or 3 digits)

ncl: '0x000000010' # NR Cell Identity (36-bit)
idLength: 32      # NR gNB ID length in bits [22...32]
tac: 1           # Tracking Area Code

linkIp: 127.0.0.1 # gNB's local IP address for Radio Link Simulation (Usually same with local IP)
ngapIp: 192.168.48.130 # gNB's local IP address for N2 Interface (Usually same with local IP)
gtpIp: 192.168.48.130 # gNB's local IP address for N3 Interface (Usually same with local IP)

# List of AMF address information
anfConfigs:
  - address: 192.168.48.131 # address of 5G_core VM ens33
    port: 38412

# List of supported S-NSSAIs by this gNB
slices:
  - sst: 1
    # sd: 1

# Indicates whether or not SCTP stream number errors should be ignored.
ignoreStreamIds: true

~
~
~
```



mcc, mnc are set to 999 and 70 each. ngapIp and gtpIp are set to ens33IP address of 5G_Cellular VM, 192.168.48.130.

```
# IMSI number of the UE, IMSI = [MCC|MNC|MSISDN] (In total 15 digits)
supi: 'imsi-999702022171063'
# Mobile Country Code value of HPLMN
mcc: '999'
# Mobile Network Code value of HPLMN (2 or 3 digits)
mnc: '70'
# SUCI Protection Scheme : 0 for Null-scheme, 1 for Profile A and 2 for Profile B
protectionScheme: 0
# Home Network Public Key for protecting with SUCI Profile A
homeNetworkPublicKey: '5a8d38864820197c3394b92613b20b91633cbd897119273bf8e4a6f4eec0a650'
# Home Network Public Key ID for protecting with SUCI Profile A
homeNetworkPublicKeyId: 1
# Routing Indicator
routingIndicator: '0000'

# Permanent subscription key
key: '465B5CE8B199B49FAA5F0A2EE238A6BC'
# Operator code (OP or OPC) of the UE
op: 'E8ED289DEBA952E4283B54E88E6193CA'
# This value specifies the OP type and it can be either 'OP' or 'OPC'
opType: 'OPC'
```

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supi is set to 999702022171063 which is the subscriber profile information set previously.